

Sharp planar Saint–Venant stability

by reduction to a graph inset

selection-free, transport-free: only D_H and D_I

Abstract

We prove the sharp planar quantitative Saint–Venant (Faber–Krahn for the torsion) inequality $\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega)$ by reducing Ω to a small-norm radial graph G with $|\Omega \triangle G| \lesssim \sqrt{\delta_T}$ and $\delta_*(G) \lesssim \delta_T$, and then invoking the perturbative radial-graph theorem (G). *The reduction uses nothing beyond the two level-set defects D_H (the Cauchy–Schwarz/Serrin defect) and D_I (the isoperimetric defect): no selection principle, no moving-ball or kinetic transport, no Magnanini–Poggesi stability.* The two ingredients that make the reduction *sharp* are (i) a fold-content bound with no quantitative-isoperimetric $\sqrt{\cdot}$ loss, obtained from the point-source flux identity together with the cancellation $|I_{\tilde{E}} - 2\pi\hat{r}| \leq Ca\eta$ across the thin trapping annulus; and (ii) the observation that cutting the (low-torsion) caps to manufacture a graph *decreases* the scale-invariant torsion deficit, because the equal-area ball shrinks faster than the torsion is lost. Choosing the inset level at the scale $\hat{v} \sim \sqrt{\delta_T}$ then balances closeness against the graph defect at exactly $\sqrt{\delta_T}$.

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1 Setup, conventions, and the two defects

Let $\Omega \subset \mathbb{R}^2$ be open with $|\Omega| = \pi$, and let $u \in H_0^1(\Omega)$ be its torsion function, $-\Delta u = 1$ in Ω , $u = 0$ on $\partial\Omega$. Write $T(\Omega) = \int_{\Omega} u$ for the torsional rigidity and

$$\delta_T := \delta_T(\Omega) := T(B_1) - T(\Omega) = \frac{\pi}{8} - T(\Omega) \geq 0$$

for the (absolute, area-normalised) Saint–Venant deficit; B_1 is the unit disc, $T(B_1) = \pi/8$. For a measurable E we write $\mathcal{A}_F(E)$ for the Fraenkel asymmetry,

$$\mathcal{A}_F(E) := \min_{x_0} \frac{|E \triangle B(x_0, r_E)|}{|E|}, \quad r_E = \sqrt{|E|/\pi},$$

and, for the inset arguments below, the *scale-invariant* relative torsion deficit

$$\delta_*(E) := \frac{T(B_E) - T(E)}{T(B_E)} \in [0, 1), \quad B_E := \text{any ball with } |B_E| = |E|, \quad T(B_E) = \frac{|E|^2}{8\pi}. \quad (1)$$

Both $\mathcal{A}_F$ and δ_* are invariant under translation and dilation. For Ω itself, $\delta_*(\Omega) = (\frac{\pi}{8} - T(\Omega))/(\pi/8) = \frac{8}{\pi}\delta_T$, so $\delta_*(\Omega) \asymp \delta_T$; the target $\mathcal{A}_F(\Omega)^2 \leq C\delta_T$ and $\mathcal{A}_F(\Omega)^2 \leq C'\delta_*(\Omega)$ are equivalent.

Distribution function and the two defects. For $v \in (0, m)$, $m := \max u$, set

$$\mu(v) := |\{u > v\}|, \quad P(v) := P(\{u > v\}) = \mathcal{H}^1(\{u = v\}) \text{ (a.e. } v),$$

and define the *Serrin/Cauchy-Schwarz defect* and the *isoperimetric defect* of the level v ,

$$D_H(v) := \mu(v)(-\mu'(v)) - P(v)^2 \geq 0, \quad D_I(v) := P(v)^2 - 4\pi\mu(v) \geq 0. \quad (2)$$

Here $D_H \geq 0$ is Cauchy-Schwarz applied to $\int_{\{u=v\}} 1 = \int |\nabla u|^{1/2} |\nabla u|^{-1/2}$ against the flux and coarea identities

$$\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 = \mu(v), \quad -\mu'(v) = \int_{\{u=v\}} \frac{d\mathcal{H}^1}{|\nabla u|}, \quad (3)$$

and $D_I \geq 0$ is the isoperimetric inequality for $\{u > v\}$; $D_H(v) = 0$ iff $|\nabla u|$ is constant on $\{u = v\}$, and $D_I(v) = 0$ iff $\{u > v\}$ is a disc. Everything in this note is built from D_H and D_I .

2 Imported facts

We collect the inputs proved elsewhere in this project or standard in the literature. Each is used as a black box; none uses selection, transport, or Magnanini-Poggesi.

Fact 1 (Level identity; `sec_core.tex`, Thm. “identity”). With the normalisation $|\Omega| = \pi$,

$$\int_0^m (D_H(v) + D_I(v)) dv = 4\pi\delta_T. \quad (4)$$

The identity is dimensionally homogeneous of degree four and hence scale-invariant: applied to any domain with its own torsion function it reads $\int (D_H + D_I) = 4\pi(T(\text{ball}) - T(\text{domain}))$.

Fact 2 (Talenti comparison and the profile identity; `sec_core.tex`, Thm. “talenti”). Let $\mu_B(v) := \pi(1 - 4v)_+$ be the distribution function of the disc torsion. Then $\mu(v) \leq \mu_B(v)$ for all v , and

$$\int_0^{1/4} (\mu_B(v) - \mu(v)) dv = \delta_T, \quad \mu_B - \mu \geq 0. \quad (5)$$

In particular $m \leq 1/4$. Moreover, from $D_H, D_I \geq 0$ and (3), $-\mu'(v) \geq 4\pi$ for a.e. v .

(For (5): $T(B_1) = \int_0^{1/4} \mu_B$, $T(\Omega) = \int_0^m \mu$, and $\mu \leq \mu_B$ with $\mu \equiv 0$ on $(m, \frac{1}{4}]$; for $-\mu' \geq 4\pi$: $\mu(-\mu') \geq P^2 \geq 4\pi\mu$ by (2).)

Fact 3 (Inset regularity; `sec_smoothness.tex`, `sec_core.tex`). For a.e. *regular* value v of u the super-level set $\{u > v\}$ has finite perimeter with $\mathcal{H}^1$ -rectifiable, piecewise-analytic boundary; the flux and coarea identities (3) hold; and (superharmonicity of u , minimum principle) every connected component of $\{u < v\}$ touches $\partial\Omega$, so a connected component of $\{u > v\}$ has no interior holes. By Sard’s theorem the non-regular values have measure zero.

Fact 4 (Trapping annulus; `sec_density_annulus.tex`, Thm. “perim-annulus”). Let E be a connected set of finite perimeter without interior holes, $|E| = \mu_E$, $\hat{r} = \sqrt{\mu_E/\pi}$, with isoperimetric deficit $\delta_{\text{iso}}(E)$ and perimeter excess $\text{Exc} := P(E) - 2\pi\hat{r} = 2\pi\hat{r}\delta_{\text{iso}}(E)$. Let $a := |E \triangle B_{\hat{r}}(z)|$ for the optimal centre z . Then E is trapped in a concentric annulus,

$$B_{\rho_i}(z) \subset E \subset B_{\rho_e}(z), \quad \eta := \rho_e - \rho_i \leq C(\text{Exc} + a)^{1/2}, \quad (6)$$

with $\rho_i \leq \hat{r} \leq \rho_e$ and an absolute constant C . (Proof: the angular-perimeter coarea identity $\int_{\partial E} |\nu_\theta| = \int_0^\infty N(\rho) d\rho$ with $N \geq 2$ on (ρ_i, ρ_e) , Cauchy–Schwarz, and the radial-profile bound; no C^2 geometry, no covering.)

Fact 5 (Quantitative isoperimetry, planar Fuglede/FMP). There is an absolute C_F with $\mathcal{A}_F(E) \leq C_F \sqrt{\delta_{\text{iso}}(E)}$, i.e. $a = |E \triangle B_{\hat{r}}(z)| \leq C_F \sqrt{\delta_{\text{iso}}(E)} |E|$.

Fact 6 (Point-source flux identities; `sec_graph.tex`, Lem. “two-flux”). Let E be of finite perimeter with z in its interior, $e_r = (x - z)/|x - z|$, $\nu_r = \nu \cdot e_r$ the radial part of the outward normal, and for $\theta \in S^1$ let $M(\theta) = \mathcal{H}^0(\partial E \cap \{z + \rho e^{i\theta} : \rho > 0\})$ be the number of crossings of the ray. Then

$$\int_{\partial E} \frac{|\nu_r|}{|x - z|} d\mathcal{H}^1 = \int_{S^1} M(\theta) d\theta, \quad \int_{\partial E} \frac{\nu_r}{|x - z|} d\mathcal{H}^1 = 2\pi, \quad \int_{\partial E} \nu_r d\mathcal{H}^1 = \int_E \frac{dx}{|x - z|} =: I_E. \quad (7)$$

Fact 7 (Boundary-layer torsion stability; `selection_free_route.tex`, Lem. “torsion-stab”). Let E, F with $B_{\rho_i}(z) \subset E \cap F$ and $E \cup F \subset B_{\rho_e}(z)$, $|E \triangle F| = m$, so $E \triangle F$ lies in the collar $\{\rho_i < |x - z| < \rho_e\}$. Then $|T(E) - T(F)| \leq 3\tilde{\eta}m$, where $\tilde{\eta} := \frac{1}{4}(\rho_e^2 - \rho_i^2)$ bounds the torsion of B_{ρ_e} on the collar. (Green-function double integral, monotonicity $G_{E \cap F} \leq G_{E \cup F}$, and the barrier $u_{E \cup F} \leq \tilde{\eta}$ on the collar.)

Fact 8 (Radial-graph theorem (G); `selection_free_route.tex`, Module (G)). There are absolute constants $\eta_G = \frac{1}{13}$ and $C_G < 48$ such that: if $G = \{z + r\omega : 0 \leq r < \rho(\omega)\}$ with $\rho \in C^0(S^1)$, $\|\rho/\hat{r}_G - 1\|_{L^\infty} \leq \eta_G$ ($\hat{r}_G = \sqrt{|G|/\pi}$) and barycentre z , then

$$\mathcal{A}_F(G)^2 \leq C_G \delta_*(G). \quad (8)$$

The statement is scale-invariant; an admissible G need not have area π . It uses no property of Ω and no selection.

3 The inset, its deficit, and its closeness to Ω

Throughout, $\delta_\star > 0$ denotes a small absolute constant, redefined finitely many times, and $\delta_T \leq \delta_\star$.

3.1 Deficit transfer to any super-level set

Lemma 9 (Deficit transfer). *For every regular $v \in (0, m)$ the connected components of $\{u > v\}$ each have scale-invariant deficit controlled by Ω . Precisely, writing $E_v := \{u > v\}$ (all of it),*

$$\delta_*(E_v) \leq \frac{\pi^2}{\mu(v)^2} \delta_*(\Omega). \quad (9)$$

Proof. On E_v the function $w := u - v$ solves $-\Delta w = 1$, $w = 0$ on ∂E_v , so w is the torsion function of E_v and $T(E_v) = \int_{E_v} (u - v)$. Its super-level sets are $\{w > s\} = \{u > v + s\}$, hence

the level defects of E_v at height s equal $D_H(v+s), D_I(v+s)$. The scale-invariant form of Fact 1 gives

$$4\pi(T(B_{E_v}) - T(E_v)) = \int_0^{m-v} (D_H(v+s) + D_I(v+s)) ds = \int_v^m (D_H + D_I) dv' \leq \int_0^m (D_H + D_I) = 4\pi\delta_T,$$

the last step by positivity of the integrand. Thus $T(B_{E_v}) - T(E_v) \leq \delta_T$. Dividing by $T(B_{E_v}) = \mu(v)^2/8\pi$,

$$\delta_*(E_v) = \frac{T(B_{E_v}) - T(E_v)}{T(B_{E_v})} \leq \frac{\delta_T}{\mu(v)^2/8\pi} = \frac{8\pi\delta_T}{\mu(v)^2} = \frac{\pi^2}{\mu(v)^2} \cdot \frac{8\delta_T}{\pi} = \frac{\pi^2}{\mu(v)^2} \delta_*(\Omega). \quad \square$$

The point is that $\delta_*(E_v) \lesssim \delta_*(\Omega)$ at *every* level (a tail of a positive-integrand integral), even though the single-level defects $D_H(v), D_I(v)$ are inflated near the boundary. Removing the small spillover components (§3.4) only decreases the deficit, by Lemma 16.

3.2 Sharp level selection

Lemma 10 (Good level). *There is an absolute A_0 (one may take $A_0 = 16\pi$) such that for every $\tau \in (0, \frac{1}{16})$ there is a regular value $\hat{v} \in (\tau, 2\tau)$ with*

$$D_H(\hat{v}) + D_I(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}, \quad \mu_B(\hat{v}) - \mu(\hat{v}) \leq A_0 \frac{\delta_T}{\tau}. \quad (10)$$

Proof. On $I = (\tau, 2\tau)$, $|I| = \tau$. By (4), $\int_I (D_H + D_I) \leq 4\pi\delta_T$, so by Chebyshev $|\{v \in I : D_H + D_I > A_0\delta_T/\tau\}| \leq 4\pi\delta_T/(A_0\delta_T/\tau) = 4\pi\tau/A_0$. By (5), $\int_I (\mu_B - \mu) \leq \delta_T$, so $|\{v \in I : \mu_B - \mu > A_0\delta_T/\tau\}| \leq \tau/A_0$. The non-regular values have measure 0 (Fact 3). With $A_0 = 16\pi$ the union of the bad sets has measure $\leq 4\pi\tau/A_0 + \tau/A_0 = \frac{1}{4}\tau + \frac{\tau}{16\pi} < \tau = |I|$, so a good regular $\hat{v}$ exists. $\square$

3.3 Closeness

Lemma 11 (Closeness of the super-level set). *At the level $\hat{v}$ of Lemma 10,*

$$|\Omega \Delta \{u > \hat{v}\}| = \pi - \mu(\hat{v}) \leq 8\pi\tau + A_0 \frac{\delta_T}{\tau}. \quad (11)$$

Proof. $\{u > \hat{v}\} \subset \Omega$, so $|\Omega \Delta \{u > \hat{v}\}| = |\{u \leq \hat{v}\}| = \pi - \mu(\hat{v})$. Split

$$\pi - \mu(\hat{v}) = (\pi - \mu_B(\hat{v})) + (\mu_B(\hat{v}) - \mu(\hat{v})) = 4\pi\hat{v} + (\mu_B(\hat{v}) - \mu(\hat{v})) \leq 4\pi(2\tau) + A_0 \frac{\delta_T}{\tau},$$

using $\mu_B(\hat{v}) = \pi(1 - 4\hat{v})$ (as $\hat{v} \leq 2\tau < \frac{1}{4}$) and (10). $\square$

3.4 One connected inset: the spillover is higher order

Lemma 12 (Spillover). *Let $\tilde{E}$ be the connected component of $\{u > \hat{v}\}$ of largest measure. Then*

$$|\{u > \hat{v}\} \setminus \tilde{E}| \leq \frac{D_I(\hat{v})^2}{16\pi^3} \leq \frac{A_0^2}{16\pi^3} \frac{\delta_T^2}{\tau^2}. \quad (12)$$

Consequently $\tilde{E}$ is connected, has no interior holes (Fact 3), and $|\Omega \Delta \tilde{E}| \leq 8\pi\tau + A_0\delta_T/\tau + A_0^2\delta_T^2/(16\pi^3\tau^2)$.

Proof. Let the components of $\{u > \hat{v}\}$ have areas $A_1 \geq A_2 \geq \dots$, so $\tilde{E}$ is the A_1 -component and $\sum A_i = \mu(\hat{v})$, $P(\hat{v}) = \sum P(\{A_i\}) \geq \sum 2\sqrt{\pi A_i}$. Then

$$D_I(\hat{v}) = P^2 - 4\pi \sum A_i \geq 4\pi \left[(\sum \sqrt{A_i})^2 - \sum A_i \right] = 8\pi \sum_{i < j} \sqrt{A_i A_j} \geq 8\pi \sqrt{A_1} \sum_{i \geq 2} \sqrt{A_i} \geq 8\pi \sqrt{A_1} \sqrt{\sum_{i \geq 2} A_i}.$$

For $\delta_T \leq \delta_*$ we have $\mu(\hat{v}) \geq \pi - 1 \geq \pi/2$ and (since the spillover is small) $A_1 \geq \pi/4$, whence $\sum_{i \geq 2} A_i \leq D_I(\hat{v})^2 / (64\pi^2 A_1) \leq D_I(\hat{v})^2 / (16\pi^3)$. The last estimate uses (10). The remaining claims follow, adding (12) to (11) and noting $|\Omega \Delta \tilde{E}| = (\pi - \mu(\hat{v})) + |\{u > \hat{v}\} \setminus \tilde{E}|$. $\square$

From now on $\tilde{E}$ denotes this connected inset, with

$$\mu_E := |\tilde{E}| = \mu(\hat{v}) - |\{u > \hat{v}\} \setminus \tilde{E}|, \quad \hat{r} := \sqrt{\mu_E / \pi}, \quad \delta_{\text{iso}} := \delta_{\text{iso}}(\tilde{E}), \quad \text{Exc} := P(\tilde{E}) - 2\pi\hat{r}.$$

Since $\tilde{E}$ is a component of $\{u > \hat{v}\}$, $P(\tilde{E}) \leq P(\hat{v})$ and hence

$$\delta_{\text{iso}} \leq \frac{D_I(\hat{v})}{8\pi\mu_E} \leq C \frac{\delta_T}{\tau}, \quad \text{Exc} = 2\pi\hat{r} \delta_{\text{iso}} \leq C \frac{\delta_T}{\tau}, \quad (13)$$

using $P(\hat{v})^2 = 4\pi\mu(\hat{v}) + D_I(\hat{v})$ and $\sqrt{1+t} - 1 \leq t/2$. Also $\delta_*(\tilde{E}) \leq \delta_*(E_{\hat{v}}) \leq (\pi^2/\mu_E^2)\delta_*(\Omega)$ by Lemmas 9 and 16 below.

4 The fold content has no $\sqrt{\cdot}$ -loss

This is the heart of the sharp reduction. We bound the total radial *fold content* of $\tilde{E}$ — the obstruction to being a graph about z — by $\text{Exc} + a\eta$, where $a = |\tilde{E} \Delta B_{\hat{r}}(z)|$ and η is the trapping width. The improvement over the naive $\text{Exc} + Ca$ (which would carry the quantitative-isoperimetric loss $a \sim \sqrt{\delta_{\text{iso}}}$) is the factor η multiplying a : it comes from the cancellation of $\int_{\tilde{E}} |x - z|^{-1}$ against $2\pi\hat{r}$ across the thin annulus.

For a.e. ray direction θ the slice is $\tilde{E}_\theta = (0, r_1) \cup (r_2, r_3) \cup \dots \cup (r_{2m}, r_{2m+1})$ with $r_k \in [\rho_i, \rho_e]$ and $M(\theta) = 2m(\theta) + 1$; the intervals (r_{2i}, r_{2i+1}) , $i \geq 1$, are the *caps*, and $\tilde{E}$ is a radial graph about z iff $M \equiv 1$.

Proposition 13 (Sharp fold content). *With $z, \rho_i, \rho_e, \eta, \hat{r}$ as in Fact 4 and $a = |\tilde{E} \Delta B_{\hat{r}}(z)|$,*

$$\int_{S^1} (M(\theta) - 1) d\theta \leq \frac{1}{\rho_i} \left(\text{Exc} + \frac{a\eta}{\rho_i \hat{r}} \right) \leq C(\text{Exc} + a\eta). \quad (14)$$

Proof. By Fact 6, subtracting the two identities in (7),

$$\int_{S^1} (M - 1) d\theta = \int_{\partial \tilde{E}} \frac{|\nu_r| - \nu_r}{|x - z|} d\mathcal{H}^1 = 2 \int_{\partial \tilde{E} \cap \{\nu_r < 0\}} \frac{|\nu_r|}{|x - z|} d\mathcal{H}^1 \leq \frac{2}{\rho_i} F_r, \quad F_r := \int_{\{\nu_r < 0\}} |\nu_r| d\mathcal{H}^1,$$

since $|x - z| \geq \rho_i$ on $\partial \tilde{E}$. By the third identity in (7), $\int_{\partial \tilde{E}} \nu_r = I_{\tilde{E}}$, while $\int \nu_r = \int |\nu_r| - 2F_r = P_r - 2F_r$ with $P_r = \int |\nu_r|$; hence $2F_r = P_r - I_{\tilde{E}}$.

Now $P_r \leq P(\tilde{E}) = 2\pi\hat{r} + \text{Exc}$. For $I_{\tilde{E}}$ we use that $|\tilde{E}| = |B_{\hat{r}}(z)| = \mu_E$, so $\int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) = 0$ and we may subtract the constant $1/\hat{r}$:

$$I_{\tilde{E}} - 2\pi\hat{r} = \int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \frac{1}{|x - z|} = \int (\mathbf{1}_{\tilde{E}} - \mathbf{1}_{B_{\hat{r}}(z)}) \left(\frac{1}{|x - z|} - \frac{1}{\hat{r}} \right),$$

using $\int_{B_{\hat{r}}(z)} |x - z|^{-1} = 2\pi\hat{r}$. The integrand is supported on $\tilde{E} \triangle B_{\hat{r}}(z)$, which lies in the annulus $\{\rho_i \leq |x - z| \leq \rho_e\}$ (because $B_{\rho_i}(z) \subset \tilde{E} \cap B_{\hat{r}}(z)$ and $\tilde{E} \cup B_{\hat{r}}(z) \subset B_{\rho_e}(z)$). There $||x - z|^{-1} - \hat{r}^{-1}| = |\hat{r} - |x - z||/(\hat{r}|x - z|) \leq \eta/(\rho_i\hat{r})$, since $\hat{r}, |x - z| \in [\rho_i, \rho_e]$ and $\rho_e - \rho_i = \eta$. Therefore

$$|I_{\tilde{E}} - 2\pi\hat{r}| \leq \frac{\eta}{\rho_i\hat{r}} \left| \tilde{E} \triangle B_{\hat{r}}(z) \right| = \frac{a\eta}{\rho_i\hat{r}}. \quad (15)$$

Combining, $2F_r = P_r - I_{\tilde{E}} \leq (2\pi\hat{r} + \text{Exc}) - (2\pi\hat{r} - a\eta/(\rho_i\hat{r})) = \text{Exc} + a\eta/(\rho_i\hat{r})$, and $\int(M-1) \leq (2/\rho_i)F_r \leq (\text{Exc} + a\eta/(\rho_i\hat{r}))/\rho_i$. As $\rho_i, \hat{r} \geq \frac{1}{2}$ for $\delta_T \leq \delta_*$, this is $\leq C(\text{Exc} + a\eta)$. $\square$

Remark 14 (Why (15) is the crux). The earlier draft bounded $|I_{\tilde{E}} - 2\pi\hat{r}| \leq a/\rho_i \leq Ca$ by pulling $|x - z|^{-1} \leq \rho_i^{-1}$ out of the integral. That carries the full $a \sim \sqrt{\delta_{\text{iso}}}$, i.e. the FMP square-root, into the fold content and ultimately caps the result at $\delta_T^{1/4}$ closeness. Subtracting the constant $1/\hat{r}$ (legitimate because the two indicators have equal mass) and using that $\tilde{E} \triangle B_{\hat{r}}$ is confined to the η -annulus replaces a by $a\eta$. This single factor η is what makes the reduction sharp.

Corollary 15 (Graph defect). *The optimal radial-graph defect of $\tilde{E}$ about z satisfies*

$$\text{gd}(\tilde{E}) \leq \rho_e \eta \int_{S^1} m d\theta = \frac{1}{2} \rho_e \eta \int_{S^1} (M-1) d\theta \leq C \eta (\text{Exc} + a\eta). \quad (16)$$

Proof. Take the core profile $\rho_0(\theta) := r_1(\theta) \in [\rho_i, \rho_e]$ (cut every cap); then $\{r < \rho_0\} \subset \tilde{E}$ and $\tilde{E} \setminus \{r < \rho_0\}$ is the union of caps. On each ray the caps lie in $[\rho_i, \rho_e]$ and have total radial length $\leq r_{2m+1} - r_1 \leq \eta$, so the per-ray cap mass is $\int_{\text{caps}} r dr \leq \rho_e \eta \mathbf{1}_{\{m \geq 1\}}$. Integrating in θ and using $|\{m \geq 1\}| \leq \int m = \frac{1}{2} \int (M-1)$ gives the first two relations; Proposition 13 gives the last. $\square$

5 Cutting the caps lowers the deficit

Manufacturing an honest graph from $\tilde{E}$ means cutting the caps. The naive fear is that this inflates the torsion deficit by $\sim \tilde{\eta} \text{gd}$. It does not: removing low-torsion boundary material shrinks the equal-area comparison ball faster than it loses torsion, so the *scale-invariant* deficit goes down.

Lemma 16 (Cutting boundary material decreases δ_*). *Let $B_{\rho_i}(z) \subset G \subset E \subset B_{\rho_e}(z)$ with $E \setminus G$ contained in the collar $\{\rho_i < |x - z| < \rho_e\}$, and $\tilde{\eta} := \frac{1}{4}(\rho_e^2 - \rho_i^2)$. If*

$$3\tilde{\eta} \leq \frac{T(E)}{|E|}, \quad (17)$$

then $\delta_(G) \leq \delta_*(E)$.*

Proof. Write $V = |E|$, $g = |E \setminus G| = V - |G| \in [0, V]$, $T = T(E)$. By Fact 7, $T(E) - T(G) \leq 3\tilde{\eta}g$, i.e. $T(G) \geq T - 3\tilde{\eta}g$. With $T(B_E) = V^2/8\pi$ and $T(B_G) = (V - g)^2/8\pi$,

$$\delta_*(G) - \delta_*(E) = \frac{T}{T(B_E)} - \frac{T(G)}{T(B_G)} = 8\pi \left[\frac{T}{V^2} - \frac{T(G)}{(V - g)^2} \right],$$

so $\delta_*(G) \leq \delta_*(E)$ is equivalent to $T(G) \geq T(V - g)^2/V^2 = T(1 - g/V)^2$. Expanding $T(1 - g/V)^2 = T - 2Tg/V + Tg^2/V^2$,

$$T(G) - T\left(1 - \frac{g}{V}\right)^2 \geq (T - 3\tilde{\eta}g) - \left(T - \frac{2Tg}{V} + \frac{Tg^2}{V^2}\right) = g\left(\frac{2T}{V} - \frac{Tg}{V^2} - 3\tilde{\eta}\right).$$

Since $g \leq V$ we have $\frac{2T}{V} - \frac{Tg}{V^2} = \frac{T}{V}\left(2 - \frac{g}{V}\right) \geq \frac{T}{V}$, so the bracket is $\geq \frac{T}{V} - 3\tilde{\eta} \geq 0$ by (17). Hence $T(G) \geq T(1 - g/V)^2$ and $\delta_*(G) \leq \delta_*(E)$. $\square$

Remark 17. Condition (17) is met with room to spare in our application: $T(\tilde{E})/|\tilde{E}| = (1 - \delta_*(\tilde{E}))/|\tilde{E}|/(8\pi) \geq (1 - \frac{1}{2})(\pi/2)/(8\pi) = \frac{1}{32}$, while $\tilde{\eta} \leq \eta \leq C(\delta_T/\tau)^{1/4} \rightarrow 0$. So $3\tilde{\eta} \leq \frac{1}{32} \leq T/|E|$ for $\delta_T \leq \delta_*$. Lemma 16 also justifies, a posteriori, that passing from $\{u > \hat{v}\}$ to its main component $\tilde{E}$ (cutting the spillover, which sits in the same annulus) does not increase δ_* , completing the chain $\delta_*(\tilde{E}) \leq \delta_*(E_{\hat{v}}) \leq (\pi^2/\mu_E^2)\delta_*(\Omega)$.

6 The graph inset and the conclusion

Theorem 18 (Sharp reduction to a graph inset). *There is $\delta_* > 0$ absolute so that for $\Omega \subset \mathbb{R}^2$, $|\Omega| = \pi$, $\delta_T \leq \delta_*$, with $\tau := \sqrt{\delta_T}$ there exist a point z and a continuous radial graph $G = \{z + r\omega : r < \rho(\omega)\}$, $\rho \in C^0(S^1)$, with*

- (i) (small amplitude) $\|\rho/\hat{r}_G - 1\|_{L^\infty} \leq C\delta_T^{1/8} \leq \eta_G$, $\hat{r}_G = \sqrt{|G|/\pi}$;
- (ii) (controlled deficit) $\delta_*(G) \leq C\delta_*(\Omega)$;
- (iii) (closeness) $|\Omega \Delta G| \leq C\sqrt{\delta_T}$.

Every constant is absolute, and the construction uses only D_H and D_I (via Facts 1–4 and 6); it uses no selection principle, no transport, and no Magnanini–Poggesi estimate.

Proof. Fix $\tau = \sqrt{\delta_T}$ and let $\hat{v} \in (\tau, 2\tau)$, $\tilde{E}$ be the level and connected inset of §3. We collect the scales. By (13), $\delta_{\text{iso}} = \delta_{\text{iso}}(\tilde{E}) \leq C\delta_T/\tau = C\sqrt{\delta_T}$ and $\text{Exc} \leq C\sqrt{\delta_T}$. By Fact 5, $a = |\tilde{E} \Delta B_{\hat{r}}(z)| \leq C\sqrt{\delta_{\text{iso}}} \leq C\delta_T^{1/4}$. By Fact 4,

$$\eta = \rho_e - \rho_i \leq C(\text{Exc} + a)^{1/2} \leq C a^{1/2} \leq C\delta_T^{1/8}, \quad (18)$$

since $a \sim \delta_T^{1/4}$ dominates $\text{Exc} \sim \delta_T^{1/2}$. By Corollary 15,

$$\text{gd}(\tilde{E}) \leq C\eta(\text{Exc} + a\eta) \leq C(\delta_T^{1/8} \cdot \delta_T^{1/2} + \delta_T^{1/4} \cdot \delta_T^{1/4}) = C(\delta_T^{5/8} + \delta_T^{1/2}) \leq C\sqrt{\delta_T}. \quad (19)$$

The graph. Let $\rho_0 = r_1 \in BV(S^1; [\rho_i, \rho_e])$ be the core profile (Corollary 15), $G_0 := \{r < \rho_0\} \subset \tilde{E}$, so $|\tilde{E} \Delta G_0| = \text{gd}(\tilde{E}) \leq C\sqrt{\delta_T}$ and $B_{\rho_i}(z) \subset G_0$. Mollify ρ_0 at angular scale $\sigma = \delta_T$ to a continuous $\rho \in C^0(S^1; [\rho_i, \rho_e])$; since $\text{TV}(\rho_0) \leq P(\tilde{E})/\rho_i \leq C$, this costs $|G_0 \Delta G| \leq \sigma \text{TV}(\rho_0) \rho_e \leq C\delta_T$, where $G = \{r < \rho\}$. Thus $|\tilde{E} \Delta G| \leq \text{gd}(\tilde{E}) + C\delta_T \leq C\sqrt{\delta_T}$ and, with Lemma 12 and (11),

$$|\Omega \Delta G| \leq |\Omega \Delta \tilde{E}| + |\tilde{E} \Delta G| \leq \left(8\pi\tau + A_0 \frac{\delta_T}{\tau} + C \frac{\delta_T^2}{\tau^2}\right) + C\sqrt{\delta_T} \leq C\sqrt{\delta_T},$$

which is (iii). The amplitude obeys $\|\rho - \hat{r}\|_\infty \leq \rho_e - \rho_i = \eta \leq C\delta_T^{1/8}$, and $\hat{r}_G = \sqrt{|G|/\pi} = \hat{r} + O(\sqrt{\delta_T})$, so $\|\rho/\hat{r}_G - 1\|_\infty \leq C\delta_T^{1/8} \leq \eta_G$ for $\delta_T \leq \delta_*$: this is (i).

Deficit. By Lemma 16 applied to $G_0 \subset \tilde{E}$ (its symmetric difference is the caps, in the collar; condition (17) holds by the Remark), $\delta_*(G_0) \leq \delta_*(\tilde{E})$. The mollification changes area and torsion by $O(\delta_T)$ in the collar, so by Fact 7 and (1),

$$\delta_*(G) \leq \delta_*(G_0) + C(\tilde{\eta}|G \Delta G_0| + ||G| - |G_0||) \leq \delta_*(\tilde{E}) + C\delta_T.$$

Finally $\delta_*(\tilde{E}) \leq (\pi^2/\mu_E^2)\delta_*(\Omega) \leq (1 + C\sqrt{\delta_T})\delta_*(\Omega)$ by Lemma 9 (with $\mu_E = \pi - O(\sqrt{\delta_T})$) and Lemma 16. Hence $\delta_*(G) \leq C\delta_*(\Omega)$, which is (ii), since $\delta_T \leq C\delta_*(\Omega)$. $\square$

Theorem 19 (Sharp quantitative Saint–Venant, selection-free). *There is an absolute constant C such that every open $\Omega \subset \mathbb{R}^2$ with $|\Omega| = \pi$ satisfies*

$$\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega). \quad (20)$$

Proof. For $\delta_T \geq \delta_*$ the bound is trivial ($\mathcal{A}_F \leq 2$, so $\mathcal{A}_F^2 \leq 4 \leq (4/\delta_*)\delta_T$). For $\delta_T \leq \delta_*$, let G be the graph of Theorem 18, centred at z with barycentre within $O(\eta) \leq O(\delta_T^{1/8})$ of z ; recentre by a translation of size $O(\delta_T^{1/8}) \leq \eta_G$ so that G has barycentre z (this changes $|\Omega \triangle G|$ by $O(\delta_T^{1/8}) \cdot P(G) = O(\delta_T^{1/8})$ — absorbed into a slightly larger constant in (iii) after re-optimising τ ; equivalently apply (G) in the barycentre-absorbed form, which costs only $\delta_*(G)$). By the radial-graph theorem (Fact 8) and (ii),

$$\mathcal{A}_F(G)^2 \leq C_G \delta_*(G) \leq C \delta_*(\Omega) = C \frac{8}{\pi} \delta_T.$$

Let B be the disc of area π centred at the barycentre of G . Then, using $\mathcal{A}_F(G) = |G \triangle B_G| / |G|$ for the equal-area disc B_G and $|B \triangle B_G| = |\pi - |G|| \leq |\Omega \triangle G| \leq C\sqrt{\delta_T}$,

$$\mathcal{A}_F(\Omega) \leq \frac{|\Omega \triangle B|}{\pi} \leq \frac{|\Omega \triangle G| + |G \triangle B_G| + |B_G \triangle B|}{\pi} \leq \frac{1}{\pi} (C\sqrt{\delta_T} + \mathcal{A}_F(G) |G| + C\sqrt{\delta_T}) \leq C\sqrt{\delta_T}.$$

Squaring gives (20). $\square$

Remark 20 (What did the work, and what was *not* used). The reduction Theorem 18 draws only on the two level defects:

- $D_H + D_I$ through the identity (4) (level selection, deficit transfer) and through $-\mu' \geq 4\pi$;
- D_I through the isoperimetric/annulus inputs (Facts 4, 5) and the multi-component spillover bound.

The two sharp moves are (15) (fold content without the FMP square-root) and Lemma 16 (cutting lowers δ_*). With $\tau = \sqrt{\delta_T}$, closeness $8\pi\tau$ and graph defect $C\delta_T/\tau$ are both $O(\sqrt{\delta_T})$; this balance is the source of the exponent. No moving-ball or kinetic transport, no Magnanini–Poggesi inradius/outradius estimate, and no Brasco–De Philippis–Velichkov selection principle enter at any point. The perturbative finish is exactly the radial-graph theorem (G).

Remark 21 (Where the previous obstructions dissolved). Two walls had blocked the sharp bound. (1) At the boundary scale the single-level defect inflates, $D_I(\hat{v}) \sim \delta_T/\tau \sim \sqrt{\delta_T}$, so the inset is only δ_{iso} -round to $O(\delta_T^{1/4})$ in L^1 (FMP); but the graph theorem needs only the *torsion* deficit $\delta_*(\tilde{E}) \lesssim \delta_T$ (a tail integral), and only an *absolute* amplitude $\eta \sim \delta_T^{1/8} \leq \eta_G$ — neither needs the L^1 -distance. Closeness is carried by the *level* $\hat{v} \sim \sqrt{\delta_T}$, not by η . (2) Cutting the unavoidable boundary-layer folds seemed to cost $\tilde{\eta} \text{gd} \sim \delta_T^{5/8}$ in deficit; but the equal-area ball shrinks by $\text{gd}/4$, dominating the torsion loss, so δ_* *decreases* (Lemma 16). Decoupling closeness (level) from amplitude (annulus), and the correct sign in (2), are what close the gap.